17 · Results, Part 1: How Search Quality Is Measured

Before reading the results you need to know what the numbers mean. Each metric looks at the **top 10** documents returned for one retrieval trigger, then the paper averages over all test triggers.

Step 1: Recall@10, "did we find it at all?"

For each trigger: 1 if a relevant document is anywhere in the top 10, else 0. Average over triggers.

> *Baseline 0.637 means: for 63.7% of searches, the right document showed up somewhere in the top 10.*

It doesn't care whether it is 1st or 10th.

Step 2: MRR@10, "how high was it?"

Reciprocal rank = 1 / (position of the first relevant document), or 0 if none is in the top 10.

First relevant document at position	Reciprocal rank
1	1.0
2	0.5
3	0.333
10	0.1
not in top 10	0

Mean reciprocal rank averages these. MRR rewards putting the right answer near the top, which matters because an agent

usually reads only the first few results.

Step 3: nDCG@10, "how good is the whole ordering?"

DCG adds up relevance scores with a discount for lower positions: a relevant document at position r contributes $1/\log_2(r+1)$. So position 1 is worth 1.0, position 3 is worth 0.5, position 10 about 0.29. **nDCG** divides by the best possible DCG, so a perfect ordering scores 1. It is useful when there are several relevant documents.

Step 4: Toy example

Suppose the relevant document for "cat food" is "Healthy dry food for cats":

System	Its position	Recall@10	RR	DCG contribution
Teacher	1	1	1.0	1.0
Student ranks it 3rd	3	1	0.333	0.5

Same recall, very different MRR. This is why the paper reports all three.

Step 5: Is the difference real? (statistics)

Test sets are finite, so some difference between two systems can be luck.

Tool	Question it answers	How
McNemar's test	On the same triggers, does one system succeed more often than the other beyond chance?	Look only at triggers where the systems *disagree*. Here the student won 140 and lost 206. $\chi^2 = (\lvert 140 - 206 \rvert - 1)^2/(140 + 206) = 12.21$, which gives $p = 0.0005$. So the student is genuinely a little worse, not just unlucky.
Bootstrap 95% CI	What range of gaps is plausible?	Resample the test triggers 1,000 times with replacement, recompute the gap each time, and take the middle 95%. For Recall@10 the gap is -3.0 points with CI $[-4.7, -1.4]$. The whole range is below zero, which agrees with McNemar.
Win / tie / loss	Where exactly do they differ?	Per trigger: 140 student-only successes, 206 baseline-only, 1,843 the same (both succeed or both fail). Agreement = 1843 / 2189 = 84.2%.

Step 6: Latency p50

The **median** time per query (half of queries are faster, half slower). The median is used because a few unusually slow queries would distort an average.